Regression Analysis

Journey Through Regression Analysis

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What is Regression Analysis ?

Core Definition

Regression analysis is a statistical technique used to **quantify and model the relationship between a dependent variable and one or more independent variables**, enabling prediction and inference.

Our Research Question

$$pr = f(mrso, mrros, evspsbl)$$

Can soil moisture, surface runoff, and evapotranspiration predict precipitation in the Mahanadi Basin?

Building on Causal Analysis

- Causal analysis established **what relationships exist** and their direction
- Moves from *"does a relationship exist?"* → *"exactly how strong and predictive is it?"*
- Partial correlation confirmed mrros → pr as strongest direct link (r = 0.935)
- Bayesian Network confirmed pr as root driver with arc strength 1.00
- Regression now **quantifies and formalises** these relationships

Why we do Regression Analysis ?

Prediction

- Build models using historical data
- Predict future values of a variable
- Evaluate performance on unseen data
- Used in forecasting (eg :- weather, finance)

Quantify Relationships

- Measures how one variable affects another
- Provides mathematical relationship (equation)
- Helps understand strength and direction
- Enables interpretation of variable influence

Validate Causal Analysis

- Supports findings from correlation analysis
- Helps identify important predictors
- Confirms relationships using data-driven models
- Improves confidence in conclusions

Real World Applications

- Climate prediction and environmental modeling
- Economic forecasting and trend analysis
- Risk assessment and decision-making
- Engineering and scientific research

How We Do Regression Analysis

From Raw Data to Validated Model

1. Collect Data *(CMIP variables)*
2. Split Data *(Time-series split)*
3. Train Regression Model
4. Predict *(Test Data)*
5. Evaluate *(R^2, RMSE, MAE)*
6. Check Assumptions *(Homoscedasticity, Normality)*

Train/Test Split

Never shuffle time series – future data cannot train the past.

TRAIN (60 %)	**TEST (40 %)**
72 months	48 months
(Jan 2026 - Dec 2031)	(Jan 2032 - Dec 2035)

Evaluation Metrics

How We Compare Models

R^2 (R - Squared)

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- Range: Generally lies from 0 to 1 (**Higher is better**)
- Proportion of variance explained
- **R^2 = 0 :** The model explains none of the variability.
- **R^2 = 1 :** The model explains all the variability.

RMSE (Root Mean Square Error)

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

- Same units as pr (mm/month)
- **Lower is better**
- Sensitive to large errors
- Squares errors → penalizes large errors heavily

MAE (Mean Absolute Error)

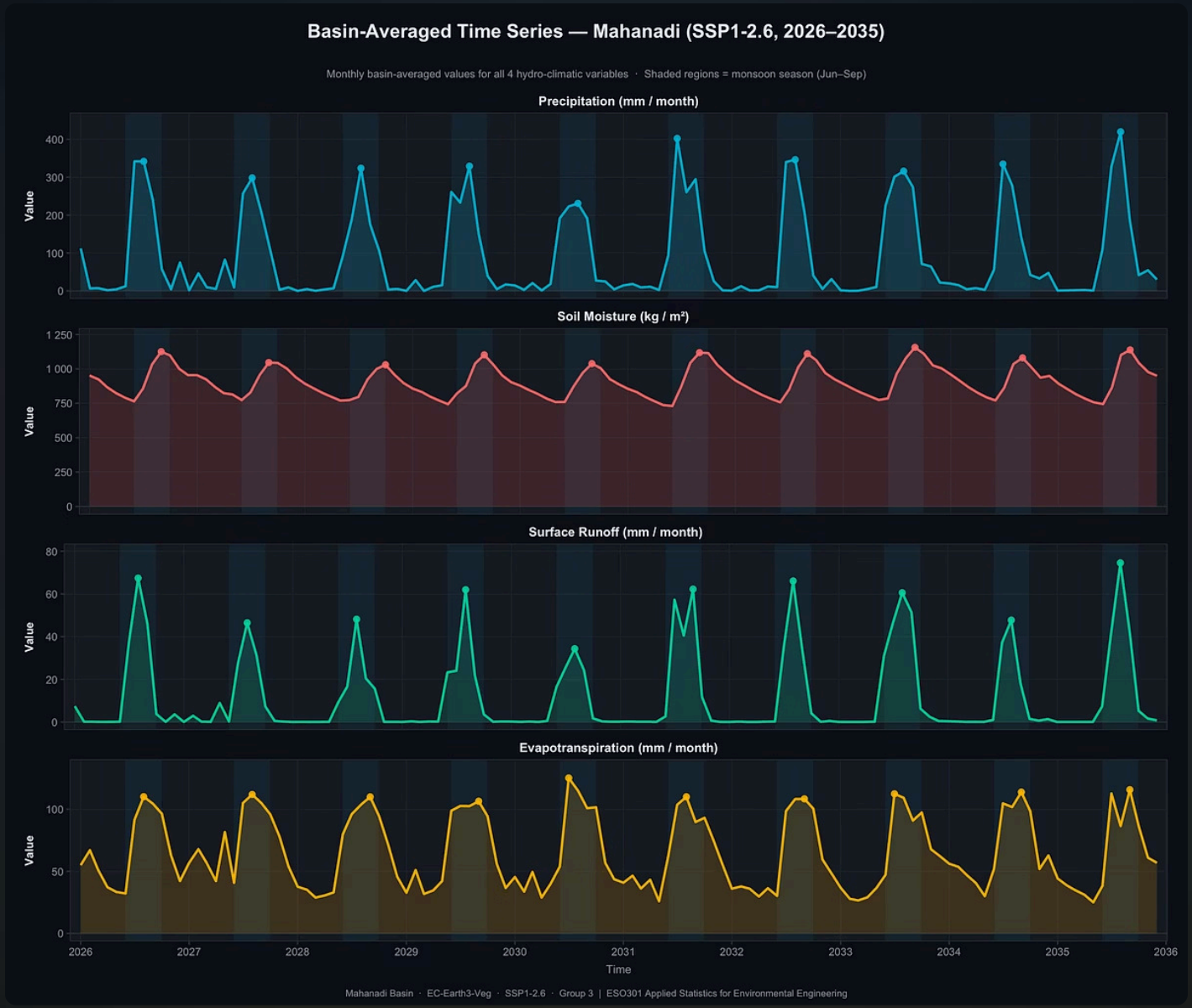
$$MAE = \frac{1}{n}\sum|y_i - \hat{y}_i|$$

- Same units as pr (mm/month)
- **Lower is better**
- Treats all errors equally
- Always lower than RMSE

- **RMSE > MAE always** – because squaring amplifies large monsoon peak prediction errors. Use both together for complete picture.
- **Why all three together:** R^2 tells you **relative** performance – what proportion of variance explained. RMSE tells you **absolute** error – how many mm/month you're actually off. MAE tells you the **typical** error without being distorted by extreme months.

What Our Models Need to Learn — Monthly Climate Patterns

120 monthly basin-averaged observations after clipping to Mahanadi shapefile



Model 1: Multiple Linear Regression (MLR)

Linear Model Equation

$$pr = \beta_0 + \beta_1(mrso) + \beta_2(mrros) + \beta_3(evspsbl)$$

What Each Term Means :-

- β_0 **(Intercept):** Value of pr when all predictors = 0.
- β_1 **(Coefficient of mrso):** For every 1 kg/m² increase in soil moisture, pr changes by β_1 mm/month.
- β_2 **(Coefficient of mrros):** For every 1 mm/month increase in runoff, pr changes by β_2 mm/month.
- β_3 **(Coefficient of evspsbl):** For every 1 mm/month increase in ET, pr changes by β_3 mm/month.
- ϵ **(Error):** What the model couldn't explain.

Actual fitted equation:

$$pr = 152.467 - 0.2342(mrso) + 4.9386(mrros) + 1.3725(evspsbl)$$

MLR : Results

Train: 2026–2031 | Test: 2032–2035 | Target: pr (Precipitation)

R^2
0.9585
Explains 95.85% of variance

RMSE
25.43 mm/month
Average prediction error

MAE
18.25 mm/month
Typical absolute error

Fitted Coefficients

Term	Value	Interpretation
β_0	152.467	Intercept
β_1	-0.2342	mrso: negative direct effect
β_2	+4.9386	mrros: dominant driver ★
β_3	+1.3725	evspsbl: moderate positive effect

mrros has the largest coefficient (+4.9386) – confirms causal analysis finding of partial r = 0.935.

VIF — Multicollinearity Check

Variance Inflation Factor | Only applicable to MLR — tree-based models are immune

What is VIF?

$$VIF_j = 1/(1 - R_j^2)$$

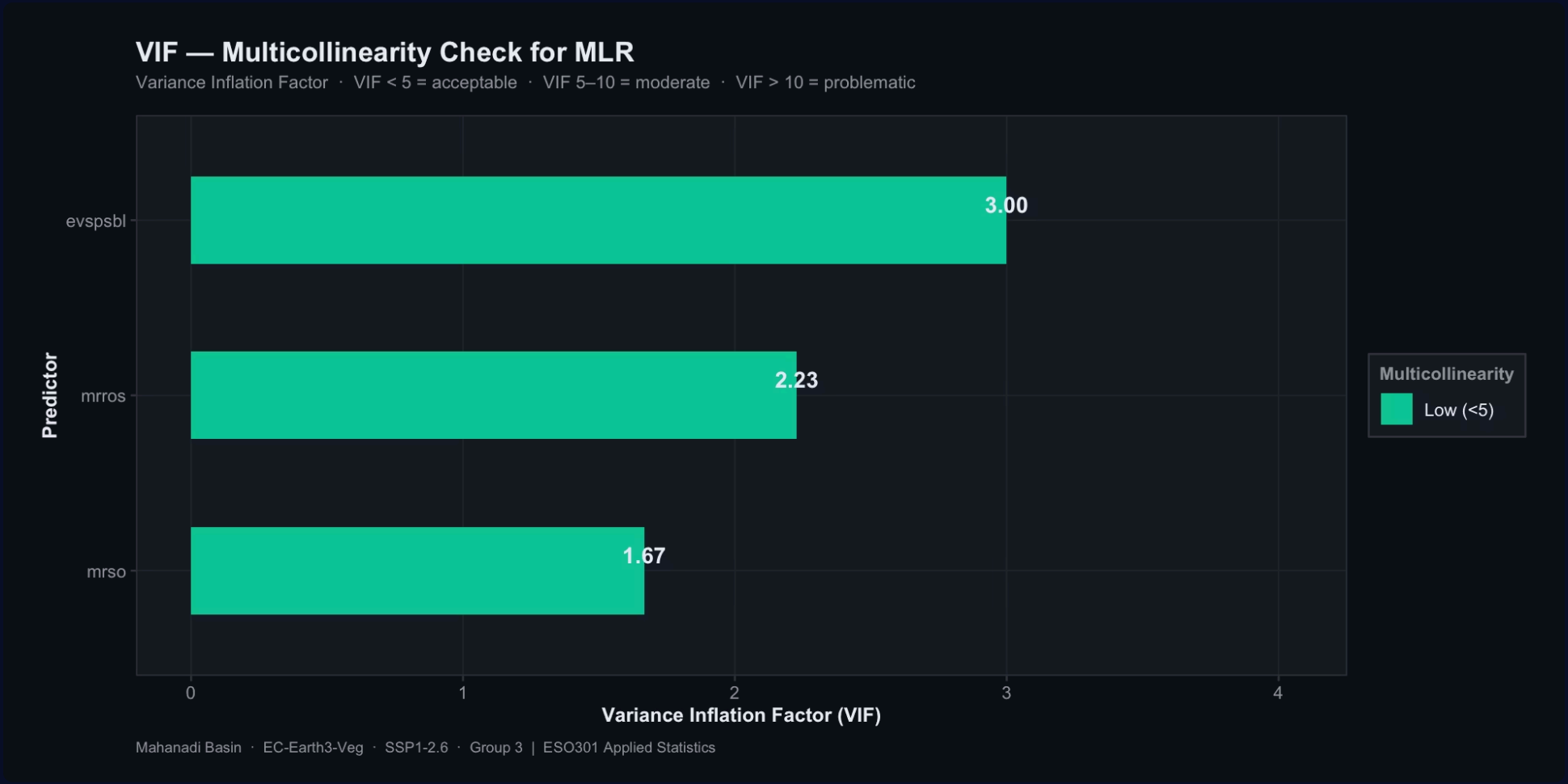
Where R_j^2 = R^2 from regressing predictor j on all other predictors

- Measures how much a coefficient is inflated due to correlation with other predictors
- Higher VIF = more correlated = less reliable coefficient
- VIF = 1 means no correlation with other predictors (ideal)

VIF Range	Status
1 – 5	Acceptable
5 – 10	Moderate
> 10	Problematic

Why VIF matters: If two predictors are nearly identical, MLR cannot separate their individual effects – coefficients become unstable. VIF < 10 confirms each predictor contributes unique information.

All VIF values well below 5 – No multicollinearity detected – MLR coefficients are statistically reliable and interpretable



MLR : Residual Diagnostics

Assumption Validation

Residuals vs Fitted

- **Linearity assumption** – whether the relationship between predictors and pr is truly linear.
- If assumption is satisfied → red line should be flat and horizontal at zero
- If assumption is violated → red line curves up or down
- The curved pattern in residuals vs fitted confirms that MLR has some linearity limitation – which directly motivates our polynomial model. This is not a failure – it is evidence that the data has non-linear structure.

Normal Q-Q Plot

- **Normality assumption** – whether the residuals follow a normal distribution.
- If assumption is satisfied → points fall on or very close to the diagonal line
- If assumption is violated → points curve away from the diagonal, especially at the tails
- Residuals are **approximately normally distributed** – the assumption is largely satisfied. The slight tail deviation is very common in climate data with extreme monsoon events and is not a serious concern for a dataset of 72 observations.

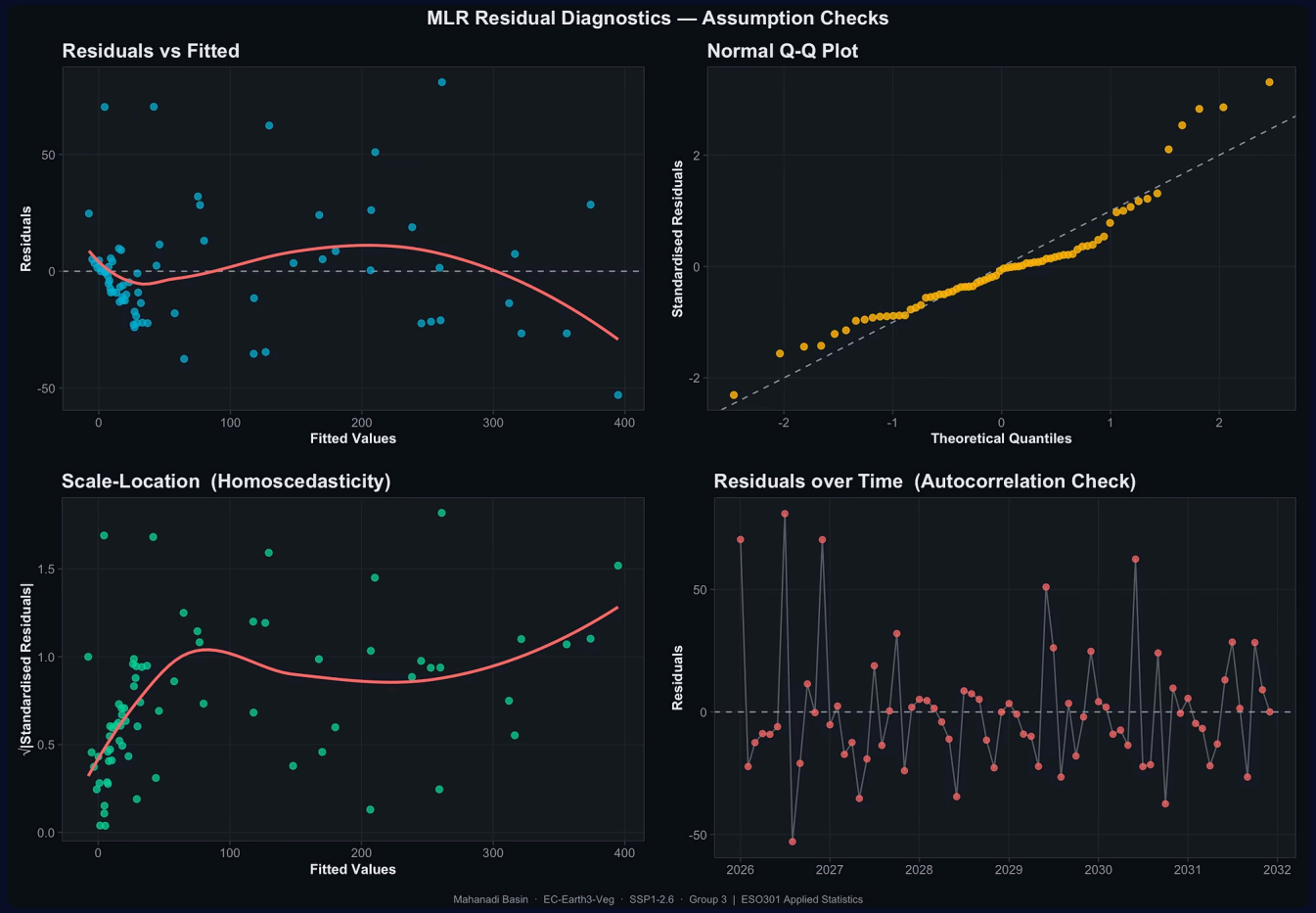
Scale-Location

- **Homoscedasticity assumption** – whether the variance of residuals is constant across all fitted values.
- If assumption is satisfied → red line should be flat and horizontal
- If assumption is violated → red line slopes up (variance increases with fitted values) – called heteroscedasticity
- The homoscedasticity assumption is **mildly violated** – variance of residuals increases at higher fitted values (the high monsoon months). This means the model is less precise during peak monsoon months than dry months.

Residuals over Time

- **Autocorrelation assumption** – whether residuals are independent over time or show a pattern.
- If assumption is satisfied → residuals scatter randomly around zero – no pattern
- If assumption is violated → systematic seasonal pattern visible – model is missing something cyclical
- No significant autocorrelation – residuals appear random over time. The large spikes correspond to extreme monsoon peak months where any linear model would struggle. The assumption is broadly satisfied.

Two assumptions fully satisfied (normality, no autocorrelation), two show mild violations at monsoon extremes (linearity, homoscedasticity) – directly motivating polynomial regression as the next model.



Model 2: Polynomial Regression

Non-Linear Model | Degree 3 | Target: pr (Precipitation)

Why MLR Fails Here

Degree 1 — MLR

Straight line only – one direction forever – cannot capture monsoon rise and fall

Degree 2 — Quadratic

One bend – can peak but cannot model asymmetric monsoon onset vs withdrawal

Degree 3 — Cubic ✓

Two bends – rise sharply, peak, fall sharply – matches Mahanadi monsoon pattern

How Degree Was Selected :-

1) Fit Degree 2 model on train data (2026–2031)

2) Fit Degree 3 model on train data (2026–2031)

3) Compare test R^2 on unseen data (2032–2035)

Degree	Test R^2	Selected
2	0.960	No
3	0.974	✓ Yes

ⓘ Mahanadi monsoon has 2 inflection points per year – rapid onset in June, rapid withdrawal in October – only degree 3 captures both

Polynomial Regression — The Equation

Degree 3 | 10 coefficients | 3 predictors each expanded to cubic

What Each Power Does :-

X^1 — Linear term
Overall slope direction – same as MLR contribution

X^2 — Quadratic term
Adds one curve – allows the relationship to peak and change direction once

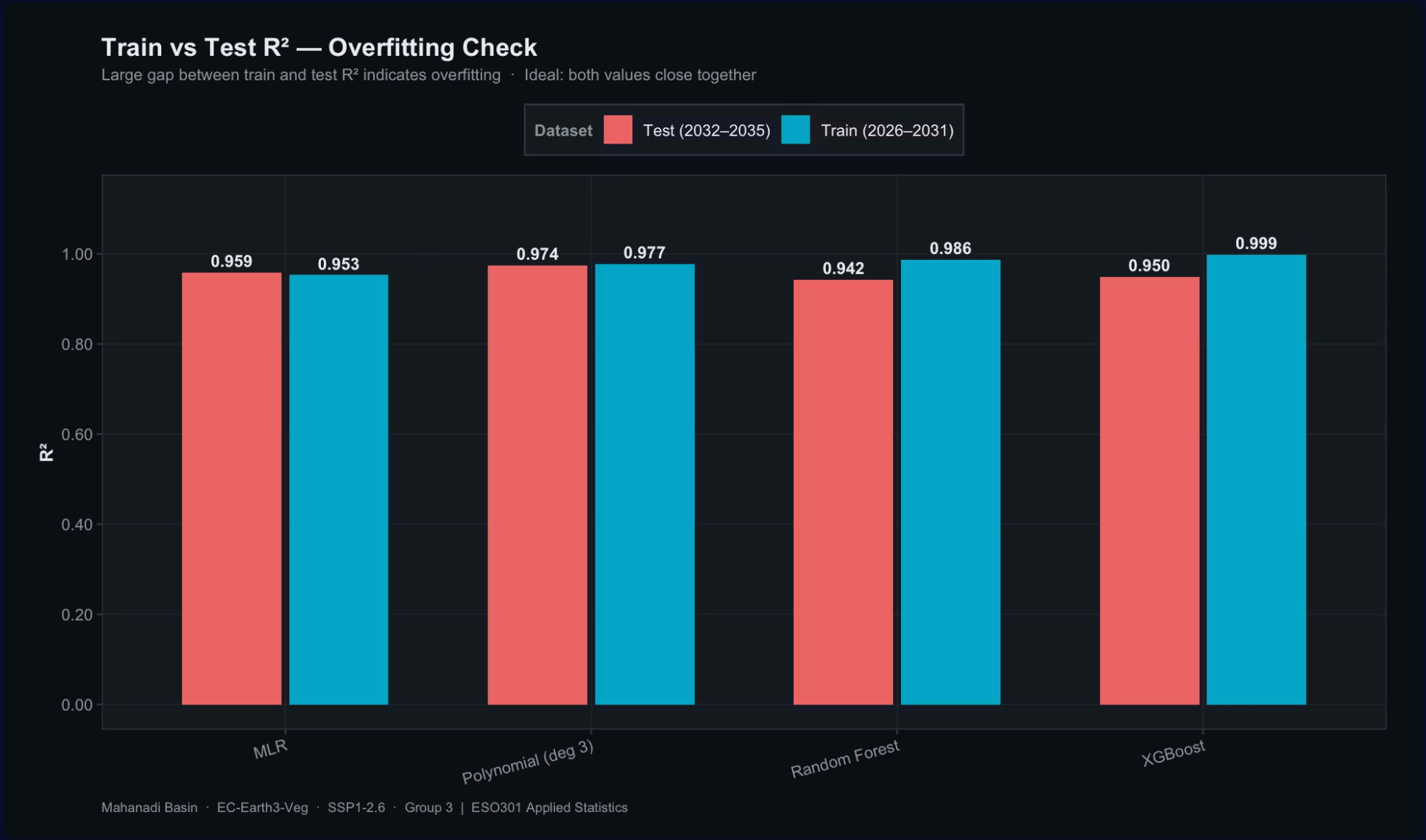
X^3 — Cubic term
Adds second bend – allows rise, peak, fall pattern – captures full monsoon cycle

Actual Fitted Coefficients from Data

Term	Variable	Value
β_0	Intercept	139.796570
β_1	$mrso^1$	-0.777820
β_2	$mrso^2$	0.00126608
β_3	$mrso^3$	-0.0000006724
β_4	$mrros^1$	11.785699 DOMINANT PREDICTOR
β_5	$mrros^2$	-0.156644
β_6	$mrros^3$	0.00090012
β_7	$evspsbl^1$	0.403710
β_8	$evspsbl^2$	0.012320
β_9	$evspsbl^3$	-0.00011381

10 coefficients vs MLR's 4 – greater flexibility captures non-linear monsoon patterns – fitted using Ordinary Least Squares on 72 training months

Despite 10 coefficients, polynomial avoids overfitting – train R^2 and test R^2 remain close, confirmed by overfitting check.


Train vs Test R² — Overfitting Check
Large gap between train and test R² indicates overfitting · Ideal: both values close together
Dataset
Test (2032–2035)
Train (2026–2031)
R²
1.00
0.80
0.60
0.40
0.20
0.00
0.959
0.953
0.974
0.977
0.942
0.986
0.950
0.999
MLR
Polynomial (deg 3)
Random Forest
XGBoost
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Polynomial Regression — Results

Degree 3 | Best Performing Model overall

R^2

0.974

Explains 97.4% of variance

Best model overall

RMSE

20.179 mm/month

Average prediction error

MAE

13.667 mm/month

Typical absolute error

Improvement over MLR Baseline

Model	R^2	RMSE	MAE
MLR	0.959	25.43	18.245
Polynomial	0.974	20.179	13,667
Improvement	+0.015	-5.251	-4.578

- **Polynomial assumes smooth curves – but what if the true relationship is more complex with hidden interactions? Random Forest will answer that.**

Foundations — Decision Trees & Bootstrap

Concepts underlying both Random Forest and XGBoost

- Decision Tree:- A model that makes predictions through a series of yes/no questions on predictor variables, splitting data at each node until a final predicted value is reached at the leaf.
- **mtry:-**It is a crucial hyperparameter that defines the number of predictor variables (features) randomly sampled as candidates at each node split when growing a decision tree.
- Bootstrap :- Randomly drawing n samples from a dataset of size n with replacement – some observations appear multiple times, some not at all – simulating multiple datasets from one.

Why These Two Concepts Matter

Used in Random Forest as:

- Bootstrap → 500 different training samples → 500 diverse trees
- Decision trees → each sample trains one tree
- Average of 500 trees → robust prediction

Used in XGBoost as:

- Decision trees → each boosting round builds one tree
- No bootstrap – uses full data
- Trees built sequentially not independently
- Each tree corrects previous tree's errors

Understanding these two foundations is essential to interpreting variable importance and why no closed-form equation exists for either model.

Model 3: Random Forest

Ensemble Method | 500 Trees | No closed-form equation

Single Decision Tree — Problem

- Makes yes/no split predictions
- If grown deep → memorises exact training data
- Fails on new data → overfitting

OVERFITTING RISK

Random Forest — Solution

- 500 independent trees
- Each on different bootstrap sample
- Average all 500 → noise cancels

ROBUST GENERALISATION ✓

Step 1
Bootstrap Sampling

72 training months → Create 500 random samples with replacement → each sample slightly different

Step 2
Build 500 Trees

One tree per bootstrap sample. At each split → only mtry=2 random predictors considered → forces diversity between trees

Step 3
Average All

500 trees each predict → Final = Average of all 500 → Individual overfitting cancels → True signal remains

Why mtry = 2 Specifically?

If all trees always split on mrros first (strongest predictor), all 500 trees would be nearly identical – averaging identical trees gives no benefit. mtry=2 forces each tree to find different patterns → creates genuine diversity

No Equation — By Design

Random Forest prediction: $\hat{y} = (1/500) \Sigma T_t(\text{mrso, mrros, evspsbl})$ where T_t = prediction of tree t. 500 nested if-else rules cannot be written as single formula. Variable importance replaces coefficients as the interpretability tool.

Random Forest — Results & Variable Importance

500 Trees | mtry=2 | Train: 2026–2031 | Test: 2032–2035

R^2
0.942
Explains 94.2% of variance
Lower than Polynomial 0.974

RMSE
29.981 mm/month
Average prediction error

MAE
16.419 mm/month
Typical absolute error

Third Confirmation of mrros

Causal Analysis
Partial r = 0.935
mrros strongest direct link

MLR Coefficient
β_2 = +4.9386 (linear)
Largest coefficient

Random Forest
Highest %IncMSE
Most important feature ✓

THREE INDEPENDENT METHODS – SAME ANSWER – MRROS DOMINANCE IS NOT A COINCIDENCE

> **Why RF Scored Lower Than Polynomial**
>
> RF R^2=0.942 vs Polynomial R^2=0.974 – Under SSP1-2.6 low emissions, Mahanadi Basin follows smooth regular monsoon cycles. Polynomial captures smooth curves efficiently – RF's tree-based complexity is unnecessary here.

Result: Random Forest R^2=0.942 — independent tree averaging is not enough. But what if trees learned sequentially from each other's mistakes? XGBoost tests exactly that.

Model 4: XGBoost

Gradient Boosting | Sequential Trees | Learning Rate $\eta = 0.05$

- **Random Forest** builds 500 trees **independently** on bootstrap samples and averages them equally – reducing variance through diversity
- **XGBoost** builds trees **sequentially** where each tree corrects errors of the previous one – reducing both variance and bias through progressive learning

Sequential Error Correction

Initial Prediction
Start with mean of training pr as baseline prediction for all observations

Compute Residuals
Calculate error for each month: residual = actual pr - predicted pr. Large residuals = model still has much to learn

Tree 1 — Fit Residuals
First tree learns to predict the residuals – not pr directly. Update: $\hat{y} = \hat{y} + \eta \times T_1$ where $\eta = 0.05$ (learning rate)

Tree 2 — Correct Remaining Errors
Residuals from Step 3 become new target → Tree 2 corrects what Tree 1 missed → Residuals shrink with each round

Early Stopping
Training stops when test error shows no improvement for 30 consecutive rounds → Optimal model selected automatically

Key Parameters

Learning rate	0.05
Max depth	4
Best iteration	300
Early stopping	30 rounds
Subsample	0.8
Col sample	0.8
Min child weight	3

XGBoost — Results & Feature Importance

Gradient Boosting | 300 Rounds | Train: 2026–2031 | Test: 2032–2035

R^2
0.9496
Explains 94.96% of variance
2nd best after Polynomial

RMSE
28.035 mm/month
Average prediction error

MAE
15.596 mm/month
Typical absolute error

Fourth Confirmation + New Insight

XGBoost Gain
0.6535 – 65.35%
Dominant feature confirmed ✓

FOUR INDEPENDENT METHODS – UNANIMOUS – FINDING IS SCIENTIFICALLY ROBUST

New Insight from XGBoost

Both RF and XGBoost agree – mrros dominates, evspsbl ranks 2nd, mrso ranks 3rd. However XGBoost reveals a much larger gap between evspsbl (30.3) and mrso (4.3) compared to RF (12.9 vs 7.7) – sequential error correction amplifies the evapotranspiration signal that independent averaging partially obscures.

XGBoost vs Random Forest

XGBoost R^2=0.9496 > RF R^2=0.9424. Sequential learning outperforms independent averaging – confirming temporal monsoon structure benefits from sequential error correction

XGBoost vs Polynomial

Polynomial R^2=0.9739 > XGBoost R^2=0.9496. Smooth curve still wins – Under SSP1-2.6 low emissions monsoon patterns are smooth and regular – polynomial captures this more efficiently than sequential tree complexity

Variable Importance — Random Forest vs XGBoost
RF: %IncMSE = % increase in error when variable is randomly permuted
XGBoost: Gain = % improvement in loss brought by each feature across all splits
Random Forest
XGBoost
Predictor Variable
mrros
evspsbl
mrso
44.7
12.9
7.7
65.4
30.3
4.3
0
10
20
30
40
50
0
20
40
60
Importance Score
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Comparative Analysis — All 4 Regression Models

All metrics computed on unseen test data (2032–2035)

Model Performance Comparison — Test Set Metrics
All metrics computed on unseen test data (2032–2035)
R² (higher is better)
RMSE (lower is better)
MAE (lower is better)
Metric Value
0.0
0.3
0.6
0.9
0
10
20
30
0
5
10
15
20
0.959
0.974
0.942
0.950
25.429
20.179
29.981
28.035
18.245
13.667
16.419
15.596
MLR
Polynomial (deg 3)
Random Forest
XGBoost
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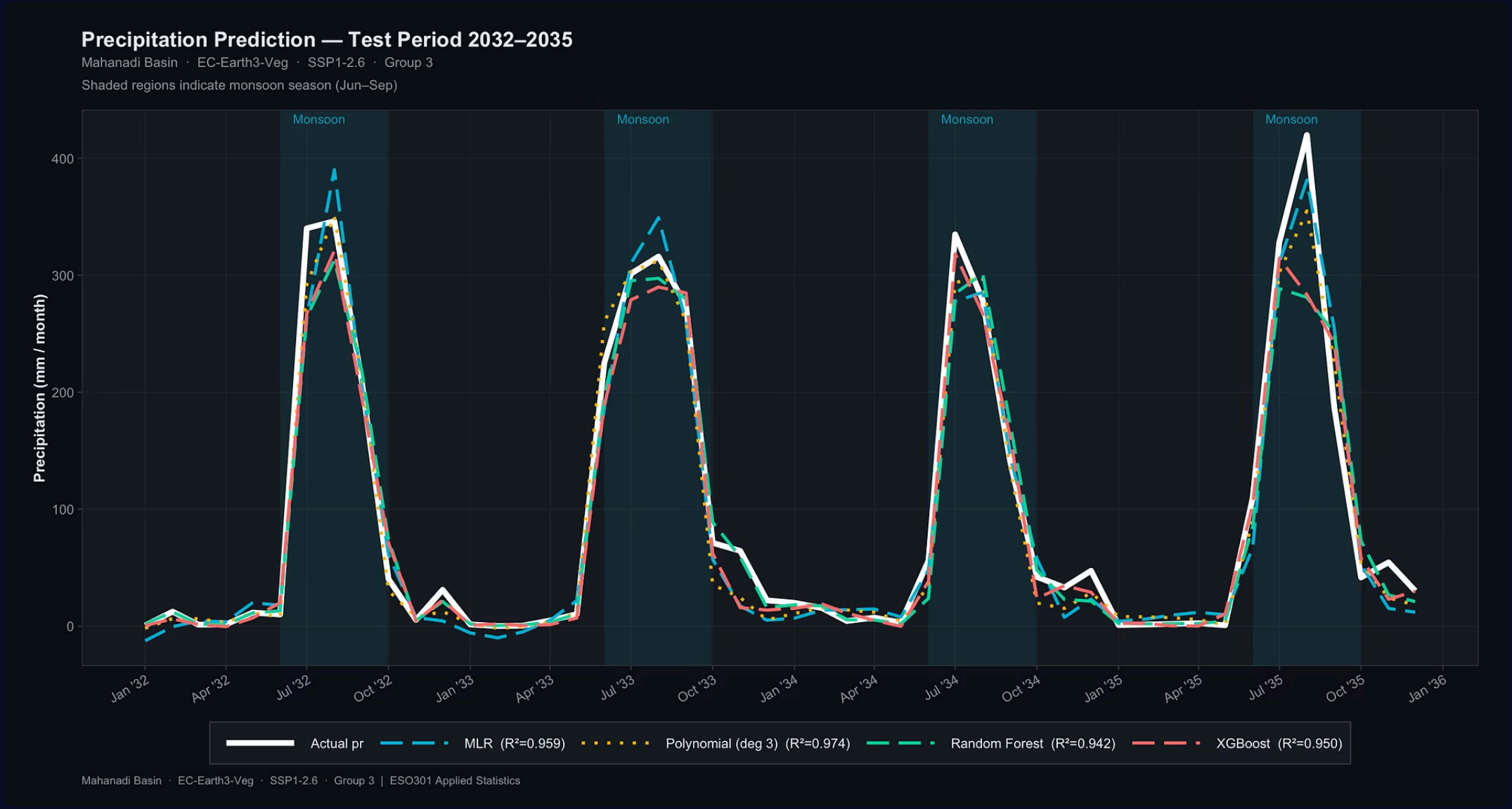
Precipitation Prediction — Test Period 2032–2035
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Shaded regions indicate monsoon season (Jun–Sep)
Monsoon
Monsoon
Monsoon
Monsoon
Precipitation (mm / month)
400
300
200
100
0
Jan '32
Apr '32
Jul '32
Oct '32
Jan '33
Apr '33
Jul '33
Oct '33
Jan '34
Apr '34
Jul '34
Oct '34
Jan '35
Apr '35
Jul '35
Oct '35
Jan '36
Actual pr
MLR (R²=0.959)
Polynomial (deg 3) (R²=0.974)
Random Forest (R²=0.942)
XGBoost (R²=0.950)
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Actual vs Predicted — Scatter Plot by Model
Dashed line = perfect prediction (slope = 1) · Closer points to diagonal = better fit
MLR
R² = 0.959
Polynomial (deg 3)
R² = 0.974
Random Forest
R² = 0.942
XGBoost
R² = 0.950
Predicted Precipitation (mm / month)
Actual Precipitation (mm / month)
0
100
200
300
400
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Split Sensitivity

Split Sensitivity — 60:40 vs 90:10 Train:Test Comparison

60:40 = Train 2026–2031 (72 months) | Test 2032–2035 (48 months)

90:10 = Train 2026–2034 (108 months) | Test 2035 only (12 months)

Similar R^2 across both splits → models are robust and not split-dependent

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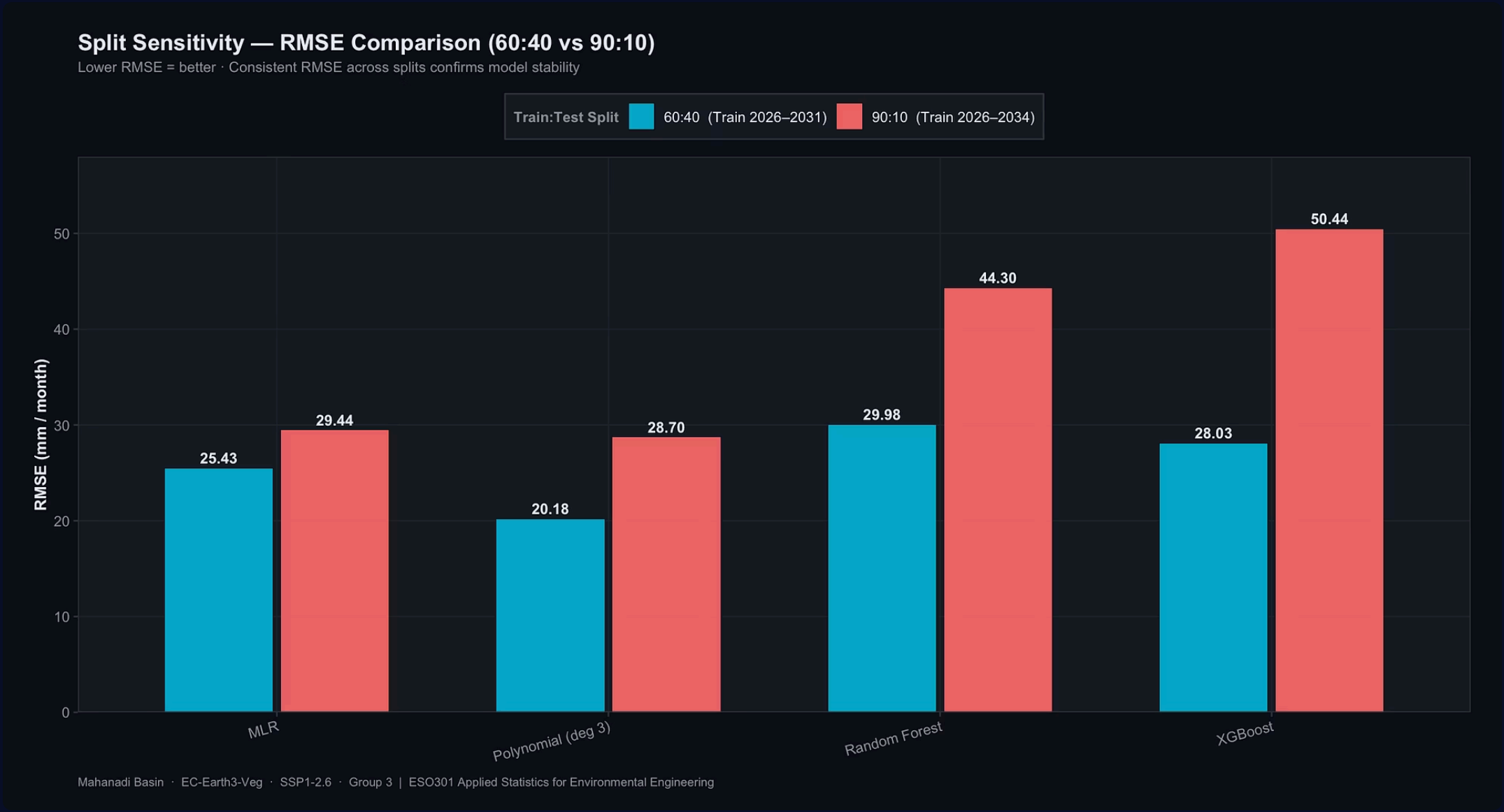
Split Sensitivity — RMSE Comparison (60:40 vs 90:10)
Lower RMSE = better · Consistent RMSE across splits confirms model stability
Train:Test Split
60:40 (Train 2026–2031)
90:10 (Train 2026–2034)
RMSE (mm / month)
0
10
20
30
40
50
25.43
29.44
20.18
28.70
29.98
44.30
28.03
50.44
MLR
Polynomial (deg 3)
Random Forest
XGBoost
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Key Findings — Regression Analysis

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Polynomial Regression (degree 3) emerged as the best model with R^2=0.974 – outperforming MLR (0.959), XGBoost (0.950) and Random Forest (0.942) – confirming that smooth non-linear monsoon patterns under SSP1-2.6 are more efficiently captured by mathematical curves than by tree-based complexity

Surface runoff (mrros) was confirmed as the dominant predictor of precipitation across all four methods – partial correlation r=0.935, largest MLR coefficient, highest RF %IncMSE and highest XGBoost Gain – convergent validity across independent frameworks makes this finding scientifically robust

Source Code in R script for Regression Analysis